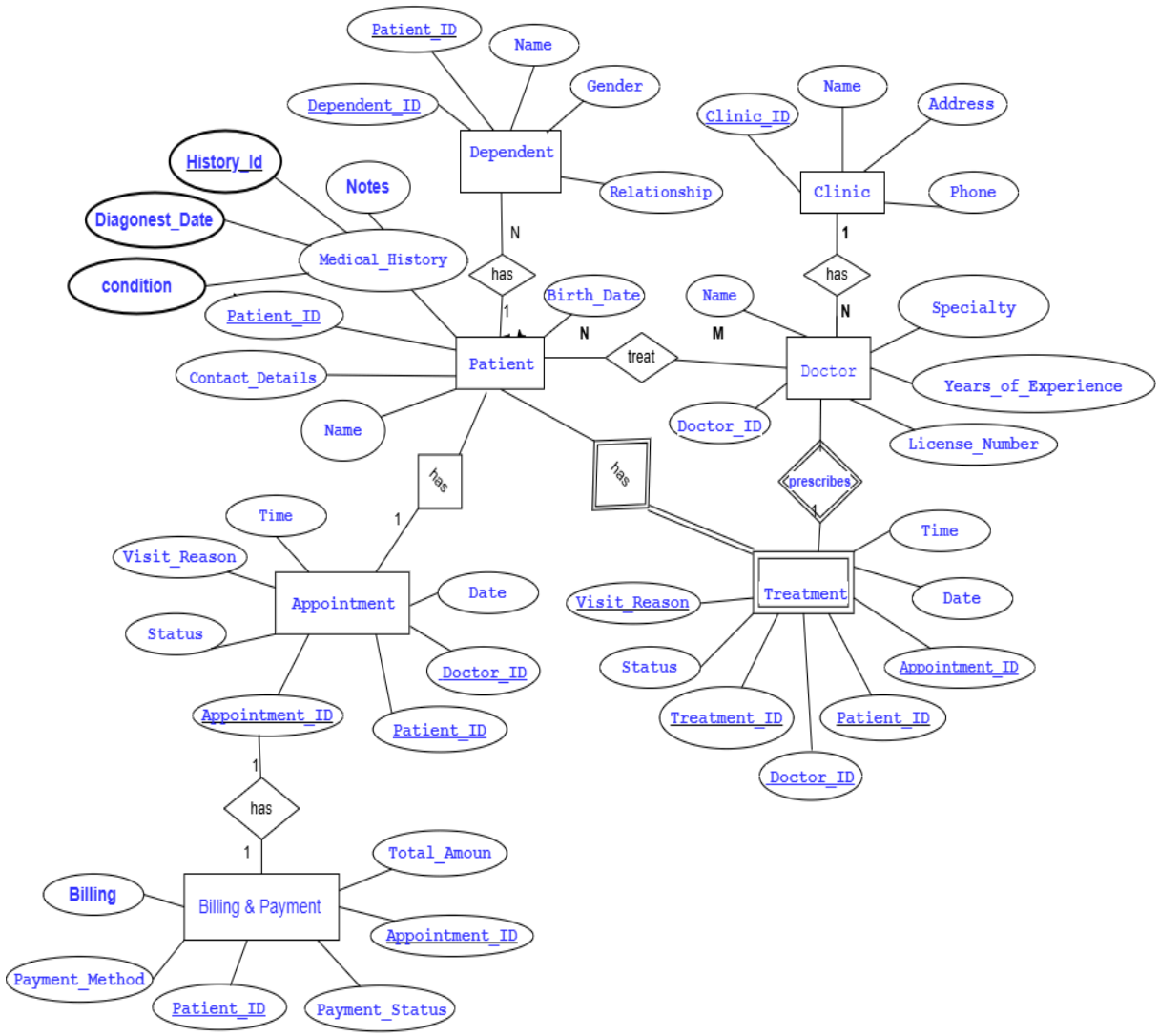
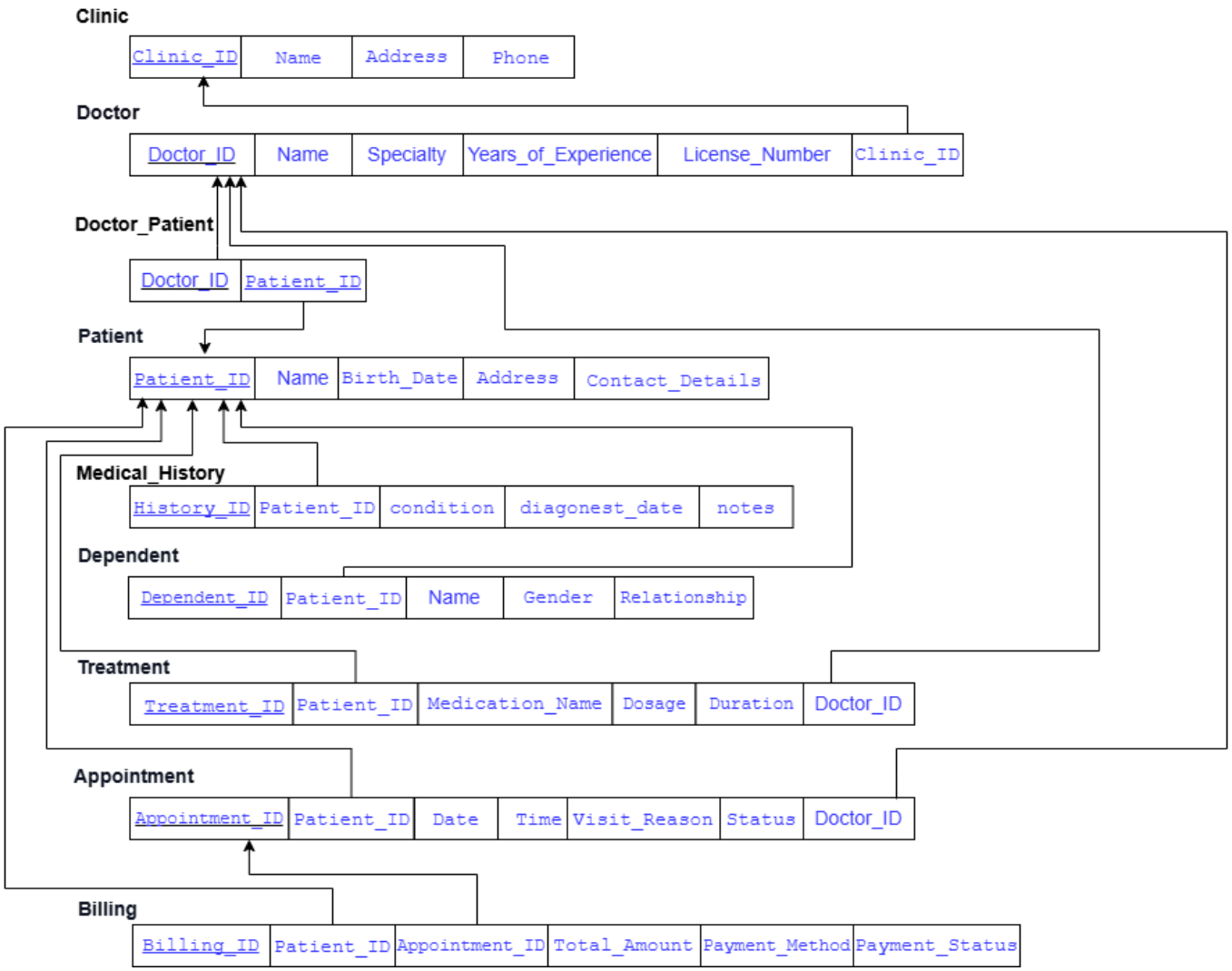
ER Diagram:


Patient_ID
Name
Gender
Dependent_ID
Dependent
Relationship
Clinic_ID
Name
Address
Clinic
Phone
History_Id
Notes
Diagonest_Date
Medical_History
condition
N
has
1
Birth_Date
Name
has
1
N
Specialty
Patient_ID
N
M
Patient
treat
Doctor
Contact_Details
Years_of_Experience
Name
Doctor_ID
License_Number
has
has
prescribes
1
Time
1
Time
Visit_Reason
Appointment
Date
Visit_Reason
Treatment
Date
Status
Doctor_ID
Status
Appointment_ID
Appointment_ID
Patient_ID
Treatment_ID
Patient_ID
Doctor_ID
1
has
1
Total_Amoun
Billing
Billing & Payment
Appointment_ID
Payment_Method
Patient_ID
Payment_Status

MAPPING of your ER diagram into Database Schema:



Tables with full specification of data types for all fields, constraint and Keys :

Clinic Table:

Attribute Name	Description	Data Type	Constraint
Clinic_id	Clinic Number (Primary Key)	Integer	Unique, Not Null
Name	Clinic Name	Varchar (50)	Unique, Not Null
Address	address	Varchar (100)	Unique
Phone	phone number	Varchar (30)	Unique

Clinc - Table

Table Name: Clinc Schema: clinic_schema

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
Clink_id	INT	☑	☑	☑	☐	☐	☐	☐	☐	
Name	VARCHAR(45)	☐	☑	☑	☐	☐	☐	☐	☐	
Address	VARCHAR(45)	☐	☐	☑	☐	☐	☐	☐	☐	
Phone	VARCHAR(45)	☐	☐	☑	☐	☐	☐	☐	☐	

Doctor Table:

Attribute Name	Description	Data Type	Constraint
Doctor_id	Doctor Number (Primary Key)	Integer	Unique, Not Null
Name	Doctor's Name	VARCHAR(100)	Unique, Not Null
Specialty	Medical Specialty	Varchar (50)	Not Null
Years_of_Experience	Experience in Years	Integer	Not Null
License_Number	License Number	Varchar (50)	Unique, Not Null

Clinic_ID	Clinic Assigned To	Integer	Foreign key referenced to Clinic(Clinic_id),not Allowed to be Null

Doctor - Table

Table Name: Doctor Schema: clinic_schema

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
Doctor_ID	INT	☑	☑	☑	☐	☐	☐	☐	☐	
Name	VARCHAR(100)	☐	☑	☐	☐	☐	☐	☐	☐	
Specialty	VARCHAR(50)	☐	☑	☐	☐	☐	☐	☐	☐	
Years_of_Experience	INT	☐	☑	☐	☐	☐	☐	☐	☐	
License_Number	VARCHAR(50)	☐	☑	☑	☐	☐	☐	☐	☐	
Clinic_ID	INT	☐	☑	☐	☐	☐	☐	☐	☑	

Patient Table:

Attribute Name	Description	Data Type	Constraint
Patient_id	Patient Number (Primary Key)	Integer	Unique, Not Null
Name	Patient Name	Varchar (100)	Unique, Not Null
Birth_Date	Date of Birth	Date	Not Null
Address	Patient Address	Varchar (100)	Not Null
Contact_Details	Contact Information	Varchar (30)	Not Null
Medical_History	Medical History	Text	Nullable

Patient - Table

Table Name: Patient Schema: clinic_schema

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
Patient_ID	INT	☑	☑	☑	☐	☐	☐	☐	☐	
Name	VARCHAR(100)	☐	☑	☐	☐	☐	☐	☐	☐	
Birth_Date	DATE	☐	☑	☐	☐	☐	☐	☐	☐	
Address	VARCHAR(200)	☐	☑	☐	☐	☐	☐	☐	☐	
Contact_Details	VARCHAR(45)	☐	☑	☐	☐	☐	☐	☐	☐	
Medical_History	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	

Treatment Table:

Attribute Name	Description	Data Type	Constraint
Treatment_ID	Treatment Number (Primary Key)	Integer	Unique, Not Null
Patient_ID	Patient Treated	Integer	FOREIGN KEY REFERENCES Patient(Patient_ID), NOT NULL
Doctor_ID	Doctor Prescribing	Integer	FOREIGN KEY REFERENCES Patient(Patient_ID), NOT NULL
Medication_Name	phone number	VARCHAR(100)	NOT NULL
Dosage	Dosage Details	VARCHAR(50)	NOT NULL
Duration	Treatment Duration	VARCHAR(50)	NOT NULL

Treatment - Table

Table Name: Treatment Schema: clinic_schema

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
Treatment_ID	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Patient_ID	INT	☐	☑	☐	☐	☐	☐	☐	☑	
Doctor_ID	INT	☐	☑	☐	☐	☐	☐	☐	☑	
Medication_Name	VARCHAR(100)	☐	☑	☐	☐	☐	☐	☐	☐	
Dosage	VARCHAR(50)	☐	☑	☐	☐	☐	☐	☐	☐	
Duration	VARCHAR(50)	☐	☑	☐	☐	☐	☐	☐	☐	

Appointment Table:

Attribute Name	Description	Data Type	Constraint
Appointment_ID	Appointment Number (Primary Key)	Integer	Unique, Not Null
Patient_ID	Patient Scheduled	Integer	FOREIGN KEY REFERENCES Patient(Patient_ID), NOT NULL
Doctor_ID	Doctor Assigned	Integer	FOREIGN KEY REFERENCES Patient(Patient_ID), NOT NULL
Date	Appointment Date	Date	NOT NULL
Time	Appointment Time	Time	NOT NULL
Visit_Reason	Reason for Visit	Varchar(255)	NOT NULL
Status	Appointment Status	Varchar(50)	Not Null

Appointment - Table

Table Name: Appointment Schema: clinic_schema

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
Appointment_ID	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Patient_ID	INT	☐	☑	☐	☐	☐	☐	☐	☑	
Doctor_ID	INT	☐	☑	☐	☐	☐	☐	☐	☑	
Date	DATE	☐	☑	☐	☐	☐	☐	☐	☐	
Time	TIME	☐	☑	☐	☐	☐	☐	☐	☐	
Visit_Reason	VARCHAR(250)	☐	☑	☐	☐	☐	☐	☐	☐	
Status	VARCHAR(50)	☐	☑	☐	☐	☐	☐	☐	☐	

Billing Table

Attribute Name	Description	Data Type	Constraint
Bill_ID	Bill Number (Primary Key)	Integer	Unique, Not Null
Patient_ID	Patient Scheduled	Integer	FOREIGN KEY REFERENCES Patient(Patient_ID), NOT NULL
Appointment_ID	Appointment Billed	Integer	FOREIGN KEY REFERENCES Patient(Patient_ID), NOT NULL
Total_Amount	Total Amount Due	Decimal(10,2)	Not Null
Payment_Method	Total Amount Due	Varchar(50)	NOT NULL
Payment_Status	Payment Method Used	Varchar(50)	NOT NULL

Dependent Table

Attribute Name	Description	Data Type	Constraint
Dependent_ID	Unique Identifier (Primary Key)	Integer	Unique, Not Null
Patient_ID	Patient Related To	Integer	FOREIGN KEY REFERENCES Patient(Patient_ID), NOT NULL
Name	Dependent's Name	Varchar(100)	Not Null
Gender	Gender of Dependent	Varchar(10)	Not Null
Relationship	Relationship to Patient	Varchar(50)	Not Null

Dependent - Table

Table Name: Dependent Schema: clinic_schema

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
Dependent_ID	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Patient_ID	INT	☐	☑	☐	☐	☐	☐	☐	☑	
Name	VARCHAR(100)	☐	☑	☐	☐	☐	☐	☐	☐	
Gender	VARCHAR(10)	☐	☑	☐	☐	☐	☐	☐	☐	
Relationship	VARCHAR(50)	☐	☑	☐	☐	☐	☐	☐	☐	

Doctor_Patient Table:

Attribute Name	Description	Data Type	Constraint
Appointment_ID	Appointment Number (Primary Key)	Integer	Unique, Not Null
Patient_ID	Patient Scheduled	Integer	FOREIGN KEY REFERENCES Patient(Patient_ID), NOT

			NULL
Doctor_ID	Doctor Assigned	Integer	FOREIGN KEY REFERENCES Patient(Patient_ID), NOT NULL

Doctor_Patient - Table

Table Name: Doctor_Patient Schema: clinic_schema

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
Doctor_Patient_ID	INT	☑	☑	☑	☐	☐	☐	☐	☐	
Doctor_ID	INT	☐	☑	☐	☐	☐	☐	☐	☑	
Patient_ID	INT	☐	☑	☐	☐	☐	☐	☐	☑	

Medical_History Table:

Attribute Name	Description	Data Type	Constraint
History_ID	Appointment Number (Primary Key)	Integer	Unique, Not Null
Patient_ID	Patient Scheduled	Integer	NOT NULL
Condition	Doctor Assigned	VARCHAR(200)	NOT NULL
Diagnosis_Date	Date of Diagnosis	DATE	NOT NULL
Notes	Additional Notes	TEXT	NULLABLE

Medical_History - Table

Table Name: Medical_History Schema: **clinic_schema**

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
History_ID	INT	☑	☑	☑	☐	☐	☐	☐	☐	
Patient_ID	INT	☐	☑	☐	☐	☐	☐	☐	☑	
Condition	VARCHAR(200)	☐	☑	☐	☐	☐	☐	☐	☐	
Diagnosis_Date	DATE	☐	☑	☐	☐	☐	☐	☐	☐	
Notes	TEXT(200)	☐	☑	☐	☐	☐	☐	☐	☐	

Fill Tables with 10 records:

Clinic Table:

Result Grid Filter Rows: Edit:

Clinic_ID	Name	Address	Phone
1	Fahd Medi...	123 Main St,...	+966123456789
2	Othman H...	456 Elm St, J...	+966987654321
3	Family Clinic	789 Oak St, ...	NULL
4	Flower Clinic	321 Pine St, ...	+966444444444
5	General H...	987 Maple St...	NULL
6	Dental Clinic	654 Birch St,...	+966222222222
7	Pediatric ...	345 Cedar S...	NULL
8	Orthopedi...	765 Walnut ...	+966999999999
9	Dermatolo...	543 Cherry ...	+966888888888
10	Cardiolog...	234 Olive St,...	+966777777777
123	Gada Clinic	14 hera street	0555784512
124	Ryaad Clinic	Fahd street	0555741236
125	Suadui Clinc	Abo Baker St...	0500589632
NULL	NULL	NULL	NULL

Doctor Table:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

Doctor_ID	Name	Specialty	Years_of_Experience	License_Number	Clinic_ID
1	Dr. Ahmed Ali	Cardiology	10	L12345	1
2	Dr. Sarah Khan	Pediatrics	5	L67890	2
3	Dr. Omar Johnson	Orthopedics	8	L54321	3
4	Dr. Emily Davis	Dermatology	12	L98765	4
5	Dr. Khalid Lee	General Medicine	7	L43210	5
6	Dr. Fatima Wilson	Gynecology	9	L32109	6
7	Dr. John Brown	Dentistry	6	L21098	7
8	Dr. Amal Green	Psychiatry	11	L10987	8
9	Dr. Hassan White	Ophthalmology	15	L98701	9
10	Dr. Leila Black	Neurology	13	L87654	10
NULL	NULL	NULL	NULL	NULL	NULL

Patient Table:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Ce

Patient_ID	Name	Birth_Date	Address	Contact_Details	Medical_History
1	Sara Ahmed	1990-05-15	789 Oak St, Riyadh	+966555555555	Asthma
2	Jana Fahd	1985-08-20	321 Pine St, Jeddah	+966444444444	Diabetes
3	Amira Saad	1978-11-10	987 Maple St, Dammam	+966333333333	Hypertension
4	Knooz Khaled	2000-03-05	654 Birch St, Al Khobar	+966222222222	Allergies
5	Hana Rafat	1995-07-25	345 Cedar St, Taif	+966111111111	NULL
6	Saad El-Ghamdy	1988-02-18	765 Walnut St, Medina	+966999999999	Arthritis
7	Talal Wagdy	1982-09-03	543 Cherry St, Mecca	+966888888888	Heart Disease
8	Yasser Faisal	1998-12-30	234 Olive St, Tabuk	+966777777777	Migraine
9	Khalid Hamdan	1976-04-12	123 Main St, Najran	+966666666666	Obesity
10	Basem Salah	1980-06-08	456 Elm St, Abha	+966555555555	Depression
NULL	NULL	NULL	NULL	NULL	NULL

Treatment Table

Result Grid | Filter Rows: | Edit: | Export/Import: | Wra

Treatment_ID	Patient_ID	Doctor_ID	Medication_Name	Dosage	Duration
1	1	2	Aspirin	1 tablet daily	7 days
3	3	1	Ibuprofen	2 tablets every 6 hours	14 days
4	4	3	Cetirizine	1 tablet daily	10 days
5	5	7	Metoprolol	1 tablet twice daily	3 months
6	6	3	Celecoxib	1 capsule daily	2 weeks
7	7	5	Amoxicillin	1 capsule every 8 hours	5 days
8	8	3	Sumatriptan	1 tablet as needed	Until resolved
9	9	5	Orlistat	1 capsule with meals	6 months
10	10	2	Fluoxetine	1 tablet daily	1 year
NULL	NULL	NULL	NULL	NULL	NULL

Billing Table

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

Appointment_ID	Patient_ID	Doctor_ID	Date	Time	Visit_Reason	Status
1	1	5	2023-11-01	10:00:00	Check-up	Confirmed
2	2	3	2023-11-02	14:00:00	Follow-up	Pending
3	3	1	2023-11-03	11:30:00	Consultation	Completed
4	4	2	2023-11-04	16:00:00	New Patient	Scheduled
5	5	4	2023-11-05	09:00:00	Emergency	In Progress
6	6	9	2023-11-06	13:00:00	Routine Check	Confirmed
7	7	10	2023-11-07	15:30:00	Surgery Follow-up	Completed
8	8	7	2023-11-08	17:00:00	Medication Adjustment	Scheduled
9	9	8	2023-11-09	12:00:00	Lab Results Discussion	Confirmed
10	10	5	2023-11-10	18:00:00	Therapy Session	Pending
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Dependent Table

Result Grid | Filter Rows: | Edit:

Dependent_ID	Patient_ID	Name	Gender	Relationship
1	1	Lara Ahmed	Female	Daughter
2	2	Omar Fahd	Male	Son
3	3	Nour Saad	Female	Wife
4	4	Khalid Davis	Male	Brother
5	5	Sarah Rafat	Female	Sister
6	6	Shahd El-Ghamdy	female	
7	7	Amal Wagdy	Female	Daughter
8	8	Yousef Faisal	Male	Son
9	9	Aisha Hamdan	Female	Niece
10	10	Mohammad Salah	Male	
NULL	NULL	NULL	NULL	NULL

Doctor_Patient Table

Result Grid | Filter Rows:

Doctor_Patient_ID	Doctor_ID	Patient_ID
1	1	1
2	1	2
3	2	1
4	2	3
5	3	4
6	3	5
7	4	6
8	4	7
9	5	8
10	5	9
NULL	NULL	NULL

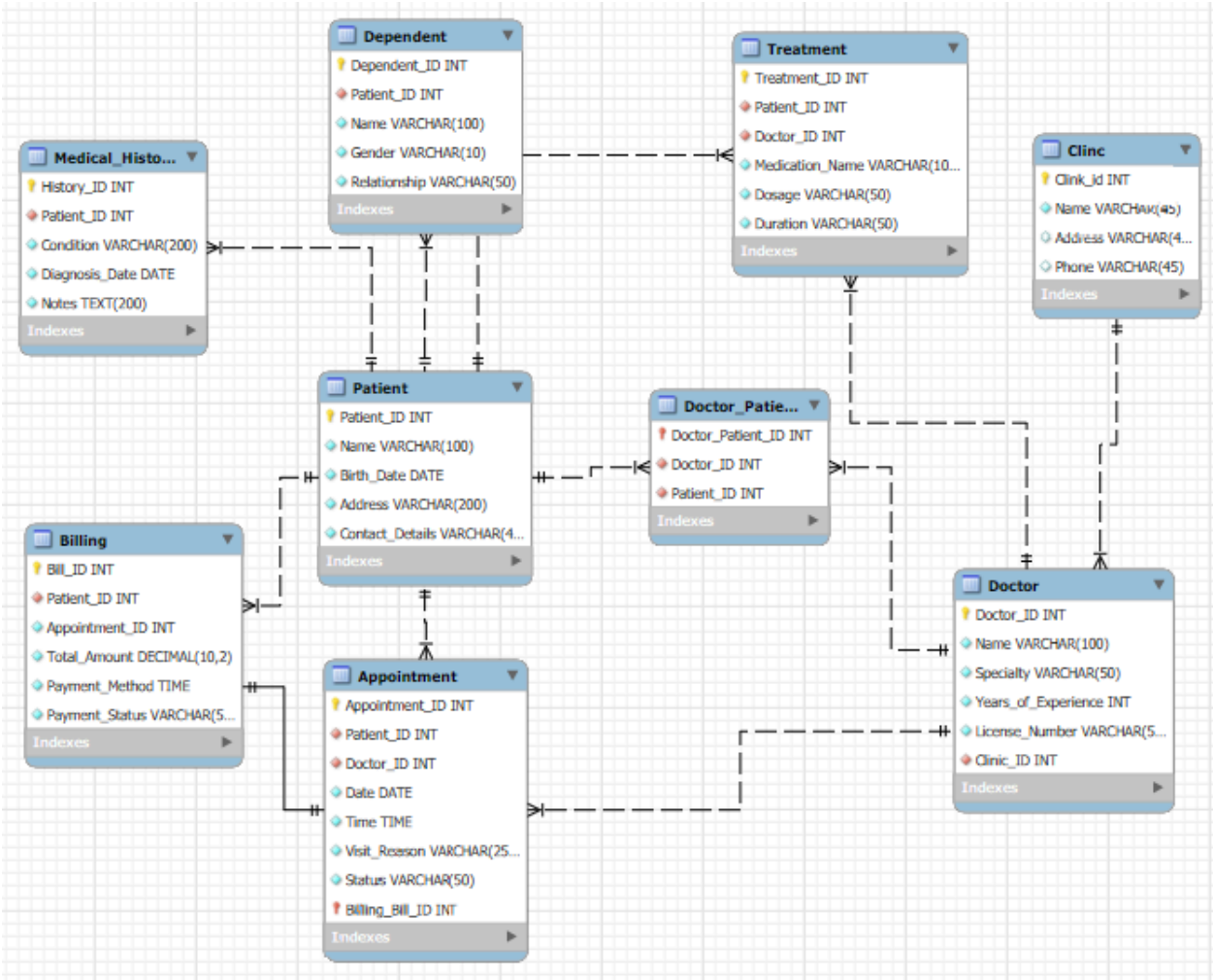
Medical_History Table:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

History_ID	Patient_ID	Conditions	Diagnosis_Date	Notes
1	1	Asthma	2020-03-15	Requires inhaler twice daily
2	1	Allergies	2019-06-20	Pollen and dust allergies
3	2	Diabetes	2018-11-10	Type 2 diabetes, managed with insulin
4	3	Hypertension	2017-08-05	Controlled with medication
5	4	Arthritis	2021-02-28	Chronic knee pain
6	5	Heart Disease	2016-05-12	Coronary artery disease
7	6	Migraine	2022-07-03	Severe headaches, triggered by stress
8	7	Obesity	2020-12-18	BMI over 35, on weight loss plan
9	8	Depression	2019-09-01	Undergoing therapy and medication
10	9	Osteoporosis	2015-03-22	Bone density decreased, calcium supplements p...
NULL	NULL	NULL	NULL	NULL

SQL Query Statements (5 marks)

❖ Database Diagram:



- **Write all SQL Queries required in your system to achieve all requirements (screenshot the result for each query).**

DML Language

1. Insert:

```
INSERT INTO Patient (Patient_ID, Name, Birth_Date, Address, Contact_Details, Medical_History)
VALUES
(11, 'Nada Ahmed', '1990-05-15', '789 Oak St, Riyadh', '+966555555555', 'Asthma');
```

10	01:24:30	INSERT INTO Patient (Patient_ID, Name, Birth_Date, Address, Contact_Details, Medical_Hi...	1 row(s) affected

11	Nada Ahmed	1990-05-15	789 Oak St, Riyadh	+966555555555	Asthma
NULL	NULL	NULL	NULL	NULL	NULL

2. Delete:

```
DELETE FROM Patient WHERE Patient_ID = 2;
```

16	01:30:08	DELETE FROM treatment WHERE Patient_ID = 2	1 row(s) affected

Treatment_ID	Patient_ID	Doctor_ID	Medication_Name	Dosage	Duration
1	1	2	Aspirin	1 tablet daily	7 days
3	3	1	Ibuprofen	2 tablets every 6 hours	14 days
4	4	3	Cetirizine	1 tablet daily	10 days

3. Update:

```
UPDATE Patient SET Address = 'New Address' WHERE Patient_ID = 1;
```

18	01:33:00	UPDATE Patient SET Address = 'Amleg Tubak' WHERE Patient_ID = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

Patient_ID	Name	Birth_Date	Address	Contact_Details	Medical_History
1	Sara Ahmed	1990-05-15	Amleg Tubak	+966555555555	Asthma

Basic Query

4. Like

```
SELECT * FROM Patient WHERE Name LIKE '%Ahmed%';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Patient_ID	Name	Birth_Date	Address	Contact_Details	Medical_History
1	Sara Ahmed	1990-05-15	Amleg Tubak	+966555555555	Asthma
11	Nada Ahmed	1990-05-15	789 Oak St, Riyadh	+966555555555	Asthma
NULL	NULL	NULL	NULL	NULL	NULL

5. Between

```
SELECT * FROM Doctor WHERE Years_of_Experience BETWEEN 5 AND 10;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Doctor_ID	Name	Specialty	Years_of_Experience	License_Number	Clinic_ID
2	Dr. Sarah Khan	Pediatrics	5	L67890	2
7	Dr. John Brown	Dentistry	6	L21098	7
NULL	NULL	NULL	NULL	NULL	NULL

6. IN

```
SELECT * FROM Clinic WHERE Clinic_ID IN (1, 2, 3);
```

Result Grid | Filter Rows: | Edit:

	Clinic_ID	Name	Address	Phone
▶	1	Fahd Medi...	123 Main St,...	+966123456789
	2	Othman H...	456 Elm St, J...	+966987654321
	3	Family Clinic	789 Oak St, ...	+966555555555
*	NULL	NULL	NULL	NULL

7. Order By

```
SELECT * FROM Patient ORDER BY Birth_Date DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cel

	Patient_ID	Name	Birth_Date	Address	Contact_Details	Medical_History
▶	4	knooz Khaled	2000-03-05	654 Birch St, Al Khobar	+966222222222	Allergies
	8	yasser fisal	1998-12-30	234 Olive St, Tabuk	+966777777777	Migraine
	5	Hana Rafat	1995-07-25	345 Cedar St, Taif	+966111111111	NULL
	1	Sara Ahmed	1990-05-15	Amleg Tubak	+966555555555	Asthma
	11	Nada Ahmed	1990-05-15	789 Oak St, Riyadh	+966555555555	Asthma
	6	saad el-ghamdy	1988-02-18	765 Walnut St, Medina	+966999999999	Arthritis
	2	Jana Fahd	1985-08-20	321 Pine St, Jeddah	+966444444444	Diabetes
	7	talal wagdy	1982-09-03	543 Cherry St, Mecca	+966888888888	Heart Disease
	10	Basem salah	1980-06-08	456 Elm St, Abha	+966555555555	Depression
	3	Amira Saad	1978-11-10	987 Maple St, Dammam	+966333333333	Hypertension
	9	Khalid Hamdan	1976-04-12	123 Main St, Najran	+966666666666	Obesity
*	NULL	NULL	NULL	NULL	NULL	NULL

8. Is Null

```
SELECT * FROM Dependent WHERE Relationship IS NULL;
```

	Clinic_ID	Name	Address	Phone
▶	3	Family Clinic	789 Oak St, ...	NULL
	5	General H...	987 Maple St...	NULL
	7	Pediatric ...	345 Cedar S...	NULL
*	NULL	NULL	NULL	NULL

Complex Query

9. Intersect - Except

```
FROM Patient
INTERSECT
SELECT Patient_ID
FROM Dependent
EXCEPT
(SELECT Patient_ID
FROM Dependent
WHERE Patient_ID = 5);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Patient_ID
1
2
3
4
6
7
8
9
10

10. Aggregate Functions + Group By + Having

```
SELECT Doctor_ID, COUNT(*) AS Treatment_Count
FROM Treatment
GROUP BY Doctor_ID
HAVING COUNT(*) > 5;
```

Result Grid | Filter Rows:

Doctor_ID	Treatment_Count
2	2
3	3
5	2

11. Natural Join

```
SELECT treatment.Treatment_ID, Doctor.Name
FROM treatment NATURAL JOIN Doctor;
```

Result Grid | Filter Rows:

Treatment_ID	Name
3	Dr. Ahmed Ali
1	Dr. Sarah Khan
10	Dr. Sarah Khan
4	Dr. Omar Johnson
6	Dr. Omar Johnson
8	Dr. Omar Johnson
7	Dr. Khalid Lee
9	Dr. Khalid Lee
5	Dr. John Brown

12. Outer Join

```
SELECT Patient.Name, Doctor.Name
FROM Patient LEFT OUTER JOIN Doctor_Patient ON Patient.Patient_ID =
Doctor_Patient.Patient_ID
LEFT OUTER JOIN Doctor ON Doctor_Patient.Doctor_ID = Doctor.Doctor_ID;
```

Result Grid | Filter Rows:

Name	Name
Sara Ahmed	Dr. Ahmed Ali
Sara Ahmed	Dr. Sarah Khan
Jana Fahd	Dr. Ahmed Ali
Amira Saad	Dr. Sarah Khan
knooz Khaled	Dr. Omar Johnson
Hana Rafat	Dr. Omar Johnson
saad el-ghamdy	Dr. Emily Davis
talal wagdy	Dr. Emily Davis
yasser fisal	Dr. Khalid Lee
Khalid Hamdan	Dr. Khalid Lee
Basem salah	NULL
Nada Ahmed	NULL

13,14. Exists, Nested Query

```
SELECT * FROM Patient WHERE EXISTS (
   SELECT 1 FROM Appointment WHERE Patient.Patient_ID = Appointment.Patient_ID
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Ce

Patient_ID	Name	Birth_Date	Address	Contact_Details	Medical_History
1	Sara Ahmed	1990-05-15	Amleg Tubak	+966555555555	Asthma
2	Jana Fahd	1985-08-20	321 Pine St, Jeddah	+966444444444	Diabetes
3	Amira Saad	1978-11-10	987 Maple St, Dammam	+966333333333	Hypertension
4	knooz Khaled	2000-03-05	654 Birch St, Al Khobar	+966222222222	Allergies
5	Hana Rafat	1995-07-25	345 Cedar St, Taif	+966111111111	NULL
6	saad el-ghamdy	1988-02-18	765 Walnut St, Medina	+966999999999	Arthritis
7	talal wagdy	1982-09-03	543 Cherry St, Mecca	+966888888888	Heart Disease
8	yasser fisal	1998-12-30	234 Olive St, Tabuk	+966777777777	Migraine
9	Khalid Hamdan	1976-04-12	123 Main St, Najran	+966666666666	Obesity
10	Basem salah	1980-06-08	456 Elm St, Abha	+966555555555	Depression
NULL	NULL	NULL	NULL	NULL	NULL

15. Function

```
DELIMITER //
CREATE FUNCTION GetPatientAge(Birth_Date DATE)
RETURNS INT
READS SQL DATA
BEGIN
   RETURN TIMESTAMPDIFF(YEAR, Birth_Date, CURDATE());
END //
DELIMITER ;
SELECT GetPatientAge('1990-05-15') AS Age;
```

Result Grid

	Age
▶	34

16. Stored Procedure

```
DELIMITER //
CREATE PROCEDURE GetDoctorsBySpecialty(IN specialty VARCHAR(50))
BEGIN
  SELECT * FROM Doctor WHERE Specialty = specialty;
END //
DELIMITER ;
```

	Doctor_ID	Name	Specialty	Years_of_Experience	License_Number	Clinic_ID
▶	1	Dr. Ahmed Ali	Cardiology	10	L12345	1

17. Trigger

```
DELIMITER //
CREATE TRIGGER UpdateBillingStatus
AFTER INSERT ON Appointment
FOR EACH ROW
BEGIN
```

```
  UPDATE Billing_and_Payment
  SET Payment_Status = 'Pending'
  WHERE Appointment_ID = NEW.Appointment_ID;
END //
DELIMITER ;
```

98	03:32:13	CREATE TRIGGER UpdateBillingStatus AFTER INSERT ON Appointment FOR EACH RO...	0 row(s) affected

18. View

```
CREATE VIEW patientAppointments AS
SELECT patient.Name, appointment.Date, appointment.Time
FROM patient
JOIN appointment ON patient.Patient_ID = appointment.Patient_ID;
SELECT * FROM patientAppointments;
```

Result Grid | Filter Rows:

Name	Date	Time
Sara Ahmed	2023-11-01	10:00:00
Jana Fahd	2023-11-02	14:00:00
Amira Saad	2023-11-03	11:30:00
knooz Khaled	2023-11-04	16:00:00
Hana Rafat	2023-11-05	09:00:00
saad el-ghamdy	2023-11-06	13:00:00
talal wagdy	2023-11-07	15:30:00
yasser fisal	2023-11-08	17:00:00
Khalid Hamdan	2023-11-09	12:00:00
Basem salah	2023-11-10	18:00:00